

Domestic violence charge records and ambient heat

Observational analysis of San Diego County Sheriff administrative releases, July 2018-December 2025

Analysis date: September 24, 2026. Primary analysis period: January 1, 2021-December 31, 2024.

Abstract

Objective. Estimate the association between modeled ambient temperature and daily counts of source identifiers carrying qualifying domestic violence (DV) charges, conditional on postal area, year-month and weekday.

Design, setting and outcome. Retrospective ecological analysis of three supplied San Diego County Sheriff exports: 214,741 valid charge rows and 136,313 source-scoped identifiers, including 12,478 core-DV and 15,161 expanded-DV identifiers. Source IDs are incident/arrest proxies, not verified persons or individual arrests. The analyzed population is the supplied administrative record set; it does not represent all DV or all police agencies in the county. [1,21]

Exposure and methods. A zero-inclusive ZIP-day panel for 2021-2024 contained 6,637 eligible core-DV IDs in 95 ZIP areas over 1,461 days. The designated primary Poisson log-link model estimated the count ratio per 10°F increase in ERA5-Land daily maximum temperature, with ZIP, year-month and weekday fixed effects. Confidence intervals used daily summed scores and seven-day Newey-West covariance. Eleven nonprimary specifications were examined retrospectively.

Results. The primary adjusted rate ratio was 1.029 (95% confidence interval [CI], 0.990–1.070; $p = 0.153$), equivalent to an estimated 2.9% change per 10°F (95% CI, –1.0% to +7.0%). The analytical heatwave contrast yielded a rate ratio of 1.162 (95% CI, 0.936–1.444). No secondary specification met a 0.05 threshold after Holm correction across the 11 tests.

Interpretation. Evidence for an association is inconclusive; the interval is compatible with no association and a modest increase. The 2018-2023 field Incident Date_Time is combined with Arrest Date/Time in 2024-2025; neither was verified as offense time. Incomplete 2025 coverage, selective charge ascertainment and modeled exposure constrain interpretation. The estimate is not a causal effect, population incidence rate or individual risk ratio.

Data domain	Scope
Administrative outcome	Supplied Sheriff records; charges, not convictions
Temperature	ERA5-Land modeled daily high, mean and low
Coverage limitation	Early 2025 incomplete; no 2026 arrest export supplied

1. Data sources and temporal coverage

The analysis uses three supplied CSV exports. Original files, source hashes and row-level provenance were retained. No agency data dictionary, extraction specification or confirmation of completeness was available. The files do not establish a census of records from every police agency in each municipality. Derived counts are reproducible from the preserved analysis snapshot. [1,21]

Valid charge rows by export: July 2018-December 2023, 172,420; 2024, 28,279; 2025, 14,042. Exact source filenames, counts and hashes are retained in source_manifest.csv.

Four rows lacked valid identifiers and timestamps and were excluded from analytic counts. The remaining 214,741 rows contain charges; several charges may share an identifier. There are 926 identical charge rows. Retaining these in the source audit does not increase distinct-identifier outcome counts.

Recorded year	All source IDs	Core DV IDs	Expanded DV IDs
2018	12,092	988	1,215
2019	24,760	1,947	2,331
2020	18,589	1,838	2,254
2021	19,018	1,875	2,351
2022	18,317	1,674	2,104
2023	18,096	1,682	2,019
2024	17,121	1,574	1,856
2025	8,283	900	1,029
Conflicting year	37	0	2

2018 begins July 1. The conflicting-year category contains identifiers whose charge rows disagree on year. Expanded counts are the union of core-DV and domestic-labeled order identifiers, not the sum of overlapping categories.

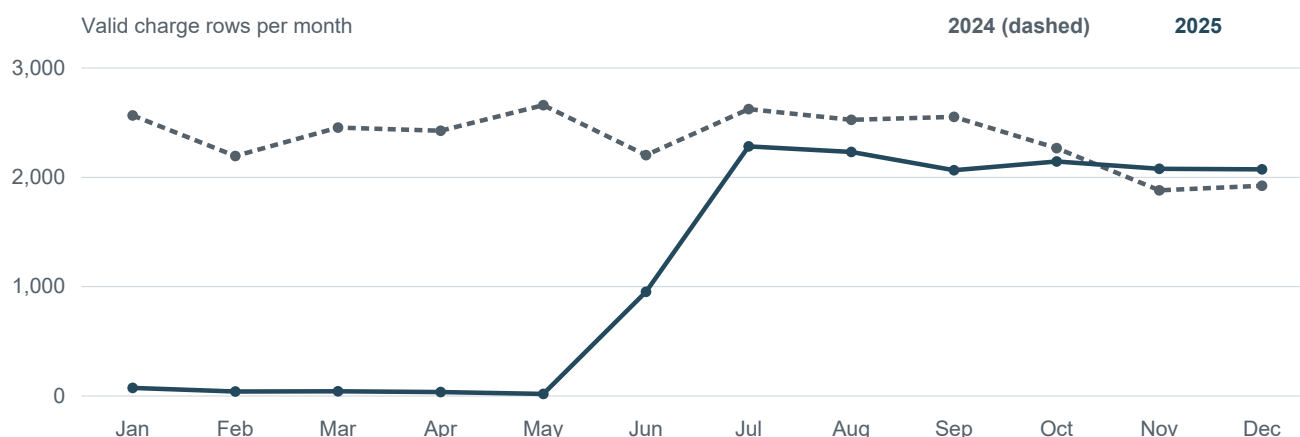


Figure 1. Monthly valid charge-row counts, 2024 and 2025. Early-2025 sparsity is a coverage discontinuity. The 2025 export has 79 dates without any supplied raw records; 83 dates have no source ID with an unambiguous recorded date. These are different audit measures. No records for 2026 were supplied.

2. Domestic violence outcome definition

Exact observed charge-type, code and description combinations were classified. Independent reconciliation covered all 1,603 combinations, with no unmapped combinations or discrepancies in inclusion tiers. Code normalization removed spaces and parentheses and retained decimal points. The primary outcome counts each source ID once if it has at least one core-DV charge. This operational definition is narrower than all family violence. [1,20,21]

Definition	Included charge families	Interpretation
Core DV	PC 243(e)(1); 273.5(a)/(f)(1)	Relationship-specific battery or corporal injury; the latter includes a prior-conviction variant. [2,3]
Core DV	Historical PC 262 variants	Recorded spousal-rape charges. AB 1171 repealed the separate section effective 2022. [5]
Core DV	PC 12022.7(e)	DV great-bodily-injury enhancement referring expressly to PC 13700(b). No second ID is added for co-charges. [4,8]
Expanded DV	PC 273.6(a)/(b); 166(c)(4)	Recorded descriptions use domestic language. Statutes include broader orders, so the relationship is not verified. [6,7]
Unconfirmed	General assault, threats, stalking, child/elder abuse, other court-order charges	Relationship or DV status cannot be established from the charge alone. Excluded unless a qualifying charge appears on the same source ID.

The core definition identified 12,487 charge lines representing 12,478 source IDs. Three DV enhancement lines co-occurred with section 273.5 and added no IDs. Two historical section 262 records, in 2018 and 2019, were additional core IDs. The expanded definition added 2,683 order-only IDs, for a total of 15,161. An unclassified charge does not establish that DV was absent.

Identifier construction and uncertainty

Identifiers were namespaced by export: Incident Number in 2018-2023 and 2024; Arrest Ref Nbr in 2025. Grouping charges prevents charge-line inflation but cannot establish people or arrest episodes. Across all data, 172 IDs have conflicting dates, 419 conflicting ZIPs and 1,637 conflicting race values. Among core IDs, none have conflicting dates; five have conflicting ZIPs. Conflicts remain flagged; no arbitrary first value determines exposure. [1,21]

The core-DV count was unchanged under checks using ID alone, ID plus timestamp, ID plus timestamp and race, and additional location/noncharge fields on qualifying DV rows. This stability supports the core count but does not validate source IDs as individual arrests. Other co-charges on a core ID can still disagree on metadata.

Limits of charge-based ascertainment

Victim relationships, case-level DV flags, narratives and dispositions were absent. General battery, assault, stalking or child/elder-abuse charges cannot establish IPV from the charge alone. Family Code 6211 also encompasses some familial relationships outside this core definition. [20] Repeal of section 262 limits consistent charge-only identification of spousal rape. [5] Charges are allegations, not adjudicated findings. The taxonomy has not been validated against case files.

3. Analytic sample and daily panel

The primary period was January 2021-December 2024. The incomplete 2025 export was excluded from the primary fit. Historical analysis covered July 2018-December 2024. July-December 2025 was evaluated separately as exploratory; apparent late-year regularity does not verify complete reporting.

Primary-sample step	Remaining core DV IDs
All core-DV source IDs in 2021-2024	6,805
One recorded date and ZIP; valid ZIP format; no warrant subtype or detention location	6,776
Exclude court categories; retain exclusively ADULT subtype	6,649
Match valid date and ZIP to the modeled weather universe	6,637

The first screening step removes 29 IDs. Court-category screening removes a further 127; no additional core IDs were lost for nonexclusive ADULT subtype among this remaining primary-period set. Twelve eligible core IDs lacked a supported ZIP-weather match. Exclusion categories should not be summed independently when they overlap.

Implemented location and warrant screens

Eligibility requires one date and five-digit ZIP, no warrant incident subtype or detention/court command/area, and an exclusively ADULT subtype. It does not exclude warrant co-charges: 67 of the 6,637 primary IDs have an explicit warrant co-charge. The source changes from Incident Date_Time in 2018-2023 to Arrest Date/Time in 2024-2025. Field equivalence and offense timing are unverified; these screens do not establish same-day violence. [1,21]

Zero-inclusive construction

The complete weather universe contains 112 postal areas and 2,741 dates. Every ZIP-date combination is retained, including dates with no qualifying records. Within the primary period, this produces 163,632 ZIP-days before model-specific all-zero-area removal. Seventeen ZIP areas had no core-DV outcome in 2021-2024 and were omitted from that fixed-effects fit to avoid divergent intercepts; they remain in the raw panel and unadjusted summaries.

Primary fitted panel	Value
ZIP areas	95
Calendar days	1,461
ZIP-days	138,795
ZIP-days with zero core-DV count	132,797
Core-DV IDs	6,637

A zero count means no qualifying record in the supplied data for that ZIP-date. It does not prove that no DV occurred or that agency reporting was complete. The 2025 gap is therefore not treated as an observed crime decline.

4. Temperature measurement and linkage

Daily high, mean and low 2-metre air temperatures were obtained from ERA5-Land reanalysis through Open-Meteo, with a fixed model choice and Fahrenheit units. Weather was requested at Census 2024 ZCTA representative internal points for observed, valid ZIP codes beginning 919, 920 or 921. This is an approximate postal-area mapping, not a verified county-polygon filter or incident-site geocode. [9-11]

There are 306,992 unique ZIP-date rows, with no missing high, mean or low values. The 112 requested postal areas share 50 returned grid coordinates, with default land-cell selection and elevation adjustment. Native resolution is approximately 9 km; these are not 112 independent weather stations. [10,25] Across all years, 12,427 of 12,478 core-DV IDs have a date/ZIP match. Matching alone does not establish model eligibility. [21]

Daily aggregation

The mean aggregates hourly modeled temperatures; it is not the midpoint of the daily high and low. Although America/Los_Angeles was requested, saved responses use fixed UTC-7: 07:00 UTC on the labeled date to before 07:00 UTC the next day. In standard time this starts at 23:00 on the preceding local date. The mismatch affects 956 of 2,741 archive dates and 513 of 1,461 primary-period dates; no May-September dates are affected. Historical daylight-saving transitions were not applied. [9,14,21]

Comparison with measured station data

Quality-filtered NOAA daily high/low records at six stations were compared with reanalysis at each station's coordinates and elevation. This is a same-site agreement comparison, not independent holdout validation. Differences include model error, spatial smoothing and observation windows; agreement cannot validate all ZIP points or personal exposure. NOAA daily averages were unavailable at all six sites; no substitute was created. [12,13,21]

NOAA station	High days	High <i>r</i>	High bias (°F)	High MAE (°F)	Low bias (°F)
San Diego airport	2,738	0.938	-1.55	2.58	-1.48
Carlsbad airport	2,713	0.900	+1.78	3.74	-0.87
Ramona airport	2,734	0.986	-4.54	4.67	+6.82
Borrego Desert Park	2,726	0.986	-3.26	3.45	-0.45
Alpine	2,740	0.982	-2.46	2.92	+1.50
Escondido No. 2	2,739	0.968	-5.82	5.92	-0.97

Station comparison. Bias is modeled minus measured temperature; MAE is mean absolute difference. Daily-high MAE was 2.58–5.92°F. Ramona daily-low bias was +6.82°F and MAE 7.16°F. Correlation does not establish agreement in absolute temperature. Source: station comparison snapshot. [21]

Absolute 80°F and 90°F indicators classify modeled temperature, not verified thermometer exceedances. Postal areas may span coastal and elevation gradients; recorded arrest locations may differ from the location of violence. Decimal precision should not be interpreted as exposure accuracy.

5. Statistical analysis

Primary estimand and count model

The primary estimand is the multiplicative change in the expected daily count of eligible core-DV source IDs associated with a 10°F increase in modeled daily maximum temperature, conditional on the specified fixed effects. The outcome is an administrative count. No population or total-arrest offset was used, so the coefficient is not an individual risk ratio or population incidence-rate ratio.

$$\log \mu_{zd} = \alpha_z + \gamma_{m(d)} + \delta_{w(d)} + \beta \frac{T_{zd}}{10}$$

$$\mu_{zd} = E[Y_{zd} | x_{zd}]$$

Y_{zd} is the core-DV count for ZIP z on date d ; T_{zd} is daily maximum temperature (°F); x_{zd} denotes model covariates. α_z includes the intercept and ZIP effect; $\gamma_{m(d)}$ and $\delta_{w(d)}$ are year-month and weekday effects; β is the temperature coefficient. ZIP effects account for stable area differences; calendar effects capture shared monthly and weekly patterns. The Poisson log link assumes a log-linear temperature association; these controls do not remove all time-varying confounding. [16]

Uncertainty estimation

Score contributions were summed across ZIPs within each date. A Bartlett-kernel Newey-West covariance used seven-day lags and the $N/(N - K)$ correction, where N is fitted ZIP-days and K is parameters. This panel covariance includes same-day and lagged cross-ZIP dependence; 138,795 rows are not treated as independent observations. Finite lags do not protect against every dependence pattern. Intervals are exponentiated normal Wald intervals (1.96 multiplier). [15,22,23]

Temperature contrasts and sensitivity analyses

This retrospective analysis designates 2021-2024 daily maximum temperature as its primary specification. There is no publicly registered protocol. Nonprimary contrasts use mean and low temperatures, maximum at least 80°F or 90°F, previous-day maximum, and a trailing three-day mean of maxima. Other fits use the expanded outcome, July 2018-December 2024, 2021-2023, and exploratory July-December 2025 data. The designation does not confer confirmatory-study status.

The analytical heatwave indicator requires at least three consecutive May-September days at or above each ZIP's 90th percentile of available May-September 2018-2024 maxima. All days in qualifying runs are flagged. The 1,010-day baseline per ZIP includes partial 2018 and the primary analysis years; it is neither an independent baseline nor a standard long-term climate normal. This definition is not an official weather alert. [21]

Multiplicity and diagnostic checks

One designated primary test was evaluated separately. Holm adjustment was applied across the original 11 nonprimary tests. Raw and adjusted p -values are reported; confidence intervals remain individual 95% intervals, not simultaneous intervals. The retrospective family designation is not a preregistered testing strategy. [17,24]

All 12 fits converged. The primary fit used 149 parameters and 10 iterations, had Pearson dispersion 1.024, and maximum absolute score 4.6×10^{-12} . Numerical coefficient and covariance calculations were checked against statsmodels on a compact data slice; maximum differences were 1.7×10^{-15} and 8.3×10^{-14} , respectively. These checks assess implementation, not validity of source definitions or causal assumptions.

6. Adjusted temperature associations

For the primary daily-maximum contrast, the rate ratio was 1.029 (95% CI, 0.990–1.070), equivalent to +2.9% per 10°F (95% CI, –1.0% to +7.0%). The interval includes the null value of 1.00 and remains compatible with a modest positive association. The analytical heatwave estimate was less precise: +16.2% (95% CI, –6.4% to +44.4%).

Specification	Period	Rate ratio (95% CI)	Raw <i>p</i>	Holm <i>p</i>	Outcome IDs
Primary: daily high +10°F	2021-24	1.029 (0.990–1.070)	0.153	-	6,637
Daily mean +10°F	2021-24	1.059 (1.004–1.117)	0.035	0.349	6,637
Daily low +10°F	2021-24	1.068 (1.011–1.127)	0.018	0.201	6,637
High ≥90°F vs <90°F	2021-24	1.087 (0.957–1.235)	0.197	1.000	6,637
High ≥80°F vs <80°F	2021-24	1.022 (0.945–1.105)	0.584	1.000	6,637
Local heatwave vs other days	2021-24	1.162 (0.936–1.444)	0.174	1.000	6,637
Previous-day high +10°F	2021-24	1.041 (1.001–1.082)	0.043	0.389	6,637
Trailing 3-day high +10°F	2021-24	1.042 (0.996–1.091)	0.073	0.585	6,637
Expanded DV: high +10°F	2021-24	1.025 (0.987–1.064)	0.208	1.000	7,825
Historical: high +10°F	Jul 2018-24	1.023 (0.993–1.054)	0.139	0.976	11,250
Exclude 2024: high +10°F	2021-23	1.021 (0.978–1.066)	0.344	1.000	5,087
Exploratory 2025: high +10°F	Jul-Dec 2025	1.065 (0.947–1.199)	0.294	1.000	827

All fitted specifications. Fixed effects and HAC7 uncertainty are as specified on page 6. The expanded-outcome row counts core and domestic-labeled order IDs. Holm correction applies to the 11 nonprimary tests; the dash denotes no multiplicity adjustment for the single primary test. No secondary result met Holm-adjusted $p < 0.05$ (minimum 0.201).

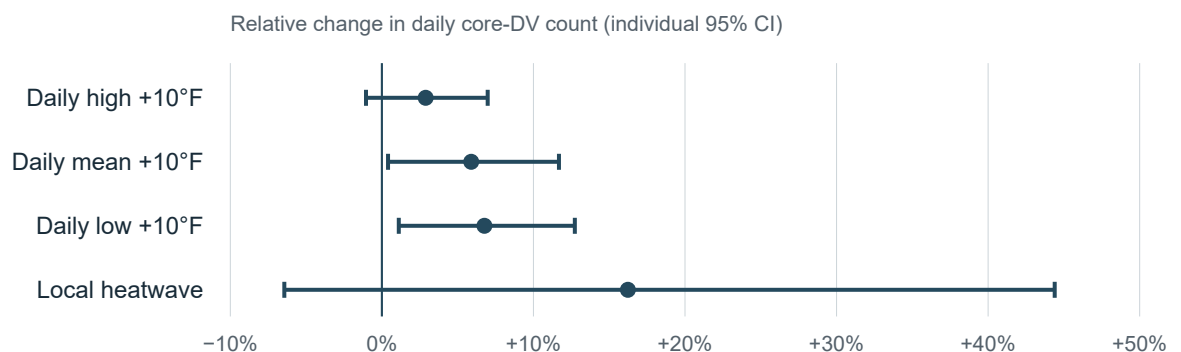


Figure 2. Selected temperature contrasts, 2021-2024. Horizontal lines are individual 95% intervals. Mean/low nominal intervals exclude no change; neither result meets the multiplicity-adjusted threshold.

7. Descriptive temporal patterns

Descriptive summaries count all core-DV source IDs with usable recorded timing in 2021-2024, before the model's location and eligibility screens. They should not be substituted for the adjusted temperature estimate. Rates divide counts by the number of corresponding calendar days, including dates with no qualifying record.

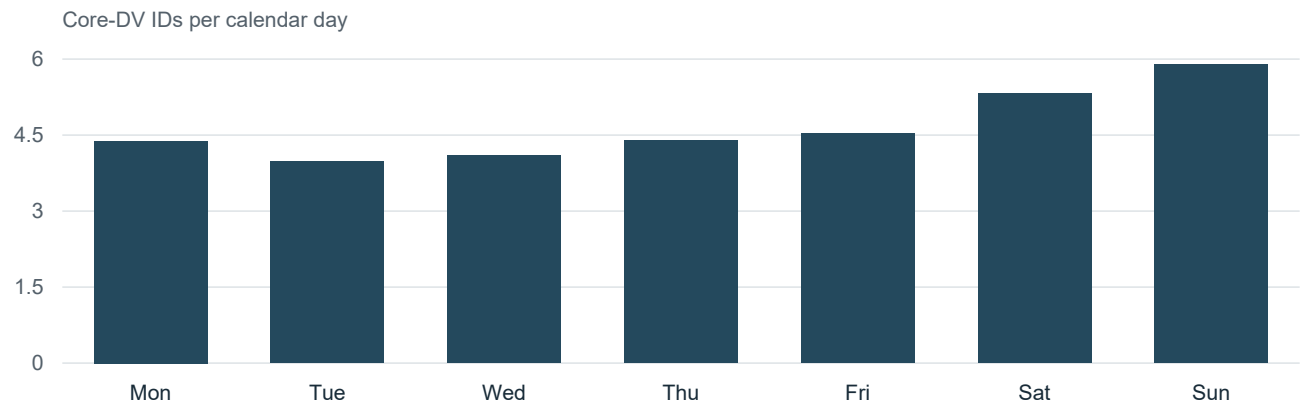


Figure 3. Recorded weekday counts per day, 2021-2024. Sunday averaged 5.89 IDs/day and Tuesday 3.98. Weekends averaged 5.61/day compared with 4.28/day on weekdays, an unadjusted difference of 31.0%. These comparisons are descriptive and not causal.

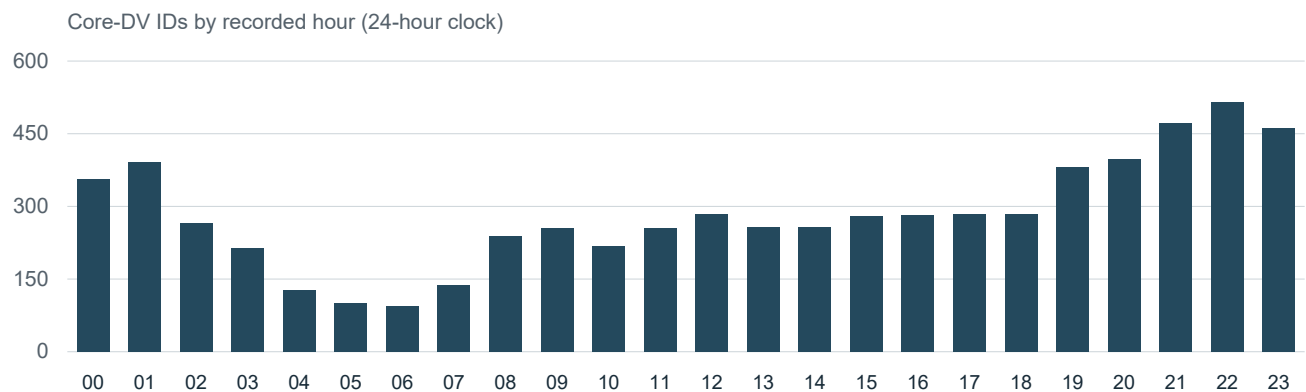


Figure 4. Recorded hour, 2021-2024. The largest bin is 22:00-22:59 (515 IDs). One core-DV ID has conflicting hour values and is not plotted. Recorded time is not verified offense time.

Seasonality and geography

July had the highest monthly daily count in 2021-2024 (5.39 core-DV IDs/day). A summer peak can reflect heat, other seasonal factors, or enforcement/reporting patterns; it is not an independent test of the heat hypothesis. The regression therefore includes year-month effects.

Across the full supplied period, the recorded city labels with the largest core-DV counts include Vista (1,727), San Marcos (1,126), Spring Valley (1,118), and the administrative label SHERIFF (829). These are source-label counts, not population rates or complete municipal totals. City=SHERIFF became more common in 2024; it cannot be interpreted as a geographic city. Complete year, city, ZIP, command, area, race, subtype and charge breakdowns are retained in the data tables.

8. Interpretation and research context

Interpretation of the primary result

The primary analysis provides insufficient evidence to establish a positive association between modeled daily maximum temperature and the count of eligible DV-charge identifiers in these records. The confidence interval is compatible with no association and with a modest increase. Statistical non-significance is not evidence of equivalence, and the results do not establish that heat has no relationship with domestic violence.

Comparison with primary empirical studies

Sanz-Barbero and colleagues analyzed May-September 2008-2016 time series in Madrid using intimate-partner femicides, police reports and helpline calls. Their heat exposure used a daily-maximum threshold of 34°C; reported associations differed by outcome and lag, appearing one to five days later. Those measures differ from charge counts indexed by recorded date and the local percentile-run definition used here. The present analysis is not a replication of their design. [18]

Zhu and colleagues studied 194,871 ever-partnered women aged 15-49 years in India, Nepal and Pakistan. They related annual mean temperature to self-reported IPV during the preceding 12 months using a cross-sectional survey design. Their exposure duration, self-reported outcome and study population differ materially from daily Sheriff administrative counts. Their reported positive association does not determine the magnitude or direction expected in this dataset. [19]

What can be compared

These studies establish that the broad question has empirical precedent. They do not provide directly interchangeable effect sizes: an annual prevalence association, delayed daily reports, and a recorded-date charge-count association estimate different quantities. A difference in statistical significance across studies is not itself evidence that their underlying effects differ. No pooled estimate, formal between-study comparison or comprehensive systematic review was performed.

Causal and population interpretation

The fitted coefficient describes a conditional association within an administrative observation process. Reporting, police contact, arrest timing, coding and export inclusion intervene between violence and the recorded outcome. Higher or lower counts may reflect changes at any of these stages. Stable ZIP effects and shared monthly/weekly controls cannot identify which stage changed. The analysis therefore cannot attribute changes to heat-induced behavior or estimate a countywide burden of violence attributable to temperature.

The expanded-outcome sensitivity evaluates an alternative charge-based ascertainment rule; it does not measure classification sensitivity or specificity. Those properties require a reference standard such as reviewed case files. Similarly, count stability under alternative grouping keys is a reproducibility check, not proof that an identifier represents one person or one arrest.

9. Limitations

Outcome selection and source completeness

The outcome depends on reporting, police contact, arrest decisions, charge coding and data extraction. It measures neither total DV occurrence nor all calls for service. Generic charges may conceal DV, and selected DV-specific charges may not result in conviction. Source identifiers can combine multiple persons or episodes. The release covers Sheriff activity rather than all agencies in the region. Continuous supplied dates before 2025 do not independently verify completeness.

Recorded date and location

The 2018-2023 export uses Incident Date_Time; 2024-2025 use Arrest Date/Time. Their equivalence was not confirmed. Timezone-naive clocks were treated as local San Diego time; that assumption was not agency-verified. Offense, report and arrest times were not separately available. Sixty-seven primary IDs retain warrant co-charges. Violence may precede the recorded date, and the daily maximum may occur after a morning record. ZIPs may describe arrest rather than offense locations. Fixed effects do not correct these timing differences. [1,21]

Exposure and model specification

Temperature is modeled at representative points, not measured at a person's location. Shared grid cells, coastal/elevation gradients, indoor conditions, air conditioning and the fixed UTC-7 boundary limit exposure accuracy. NOAA station comparisons show meaningful absolute bias. These errors may alter threshold classification and estimates; their net direction cannot be established from the available data.

ZIP, year-month and weekday controls do not adjust for every changing factor, including holidays, local seasonal patterns, service availability or police activity. The primary model assumes one common log-linear association; a post-hoc quadratic check does not establish linearity or exclude thresholds. No causal adjustment set was validated. HAC covariance addresses selected count dependence, not uncertainty from missing records, charge definitions or exposure measurement. No missing-record imputation was attempted.

Information required to extend the study

A stronger follow-up would require complete 2025 records across source systems and 2026 data; extraction filters and system-change dates; stable incident, arrest and person identifiers; victim-relationship and case-level DV flags; offense, report and arrest timestamps; location-field definitions; and explanations for the 2024 city-label change and 2025 subtype changes. Offense-date data and independently measured local weather would permit a more direct test of the proposed temporal association.

10. Reproducibility and verification

This report is a presentation of the verified analysis snapshot dated September 24, 2026. The PDF builder reads saved results; it does not refit statistical models or modify original CSV files. The analysis code, exact charge map, weather responses and file hashes form the reproducibility record. Model specification names in `model_results.csv` identify every reported fit.

Component	Reproducibility artifact
Input provenance	<code>source_manifest.csv</code> : source filenames, SHA-256 hashes and row counts
Source audit	<code>audit.json</code> ; <code>all_charge_rows_normalized.csv</code> ; <code>source_id_conflicts.csv</code>
DV taxonomy	<code>classification/charge_classification_map.csv</code> ; <code>classification/audit_classification.py</code>
Record grouping	<code>analyze.py</code> ; <code>all_source_id_groups.csv</code> ; <code>dv_related_source_id_groups.csv</code>
Weather acquisition	<code>weather/acquire_weather.py</code> ; <code>weather_provenance.json</code> ; <code>weather_locations.csv</code> ; raw/ responses
Station comparison	<code>weather/crosscheck_noaa.py</code> ; <code>noaa_crosscheck.csv</code> ; <code>noaa_provenance.json</code>
Daily analysis and fits	<code>model_heat.py</code> ; <code>daily_zip_analysis_panel.csv</code> ; <code>model_results.csv</code> ; <code>model_diagnostics.json</code>
Matched records	<code>dv_records_with_daily_weather.csv</code> : matched or explicit unavailable temperatures
Presentation	<code>DV_heat_analysis.xlsx</code> ; <code>publication/pdf/build_report.py</code>

Verification performed

Source hashes, charge/group totals, unique join keys, temperature ordering and the 6,637-ID primary sample reconcile. All original fits converged. A separate conditional-Poisson computation using only public aggregates reproduced the full primary coefficient within 2.1×10^{-16} and HAC7 standard error within 1.2×10^{-13} . Direct Holm calculation matched exported values within 1.1×10^{-16} . These computational checks are not independent human peer review. [21]

Post-hoc statistical audit

HAC14: ratio 1.0289, 95% CI 0.9891–1.0703; $p = 0.157$.

HAC28: ratio 1.0289, 95% CI 0.9879–1.0716; $p = 0.170$.

Quadratic exposure check (HAC14): curvature $p = 0.655$; 80°F versus 70°F ratio 1.031 (95% CI, 0.989–1.074); 90°F versus 80°F ratio 1.040 (95% CI, 0.973–1.112). These exploratory, unadjusted diagnostics are separate from the original 12 models and 11-test Holm family. They do not establish equivalence, linearity or model validity. [21]

Source-file identification

Arrests_Jul_2018-Dec_2023_for_Release(Arrests & Citations).csv

SHA-256: 463c0be35552d94e2e73e01e30c84f0dde9d53bb7ac5e5de0047321b1a68411a

Arrests_2024_for_Release(Sheet1).csv

SHA-256: 7d1874af6fdb635ef2cbf7bec55e956ec0cab0d6e43840dfc62f24bce76ebbb

Arrests_2025_for_Release(NetRMS Arrests 2025).csv

SHA-256: 07642b0aa5908c804088e5a933038f28edceb15c736ebad7f8ce5d846b4f3a7a

Reproduction should use the saved weather responses. Fresh API requests can return a different fixed timezone offset; compare provenance before mixing snapshots. No arrest details were submitted to weather services. The report contains aggregate results and no person-level case narratives.

11. References and data attribution

Numbered references identify legal definitions, source data, methods and contextual studies. External titles are clickable. Original source filenames and hashes appear on page 11. Current statutes were checked against recorded code descriptions; a case-specific legal adjudication was not undertaken. All online sources accessed September 24, 2026.

1. Supplied administrative data. Three Sheriff arrest CSV exports covering July 2018-2023, 2024 and 2025; `source_manifest.csv`. No agency dictionary supplied.
 2. California Penal Code 243. Subdivision (e)(1): relationship-specific battery.
 3. California Penal Code 273.5. Relationship-specific corporal injury and prior-conviction provisions.
 4. California Penal Code 12022.7. Subdivision (e): DV great-bodily-injury enhancement.
 5. California AB 1171, chaptered 2021. Repeal of separate spousal-rape section 262 and related changes.
 6. California Penal Code 273.6. Protective-order offenses, in chapter text.
 7. California Penal Code 166. Contempt and protective-order subsections.
 8. California Penal Code 13700. Law-enforcement domestic-violence relationship definition.
 9. Open-Meteo Historical Weather API. Daily variables, model selection and API options.
 10. Copernicus Climate Change Service / ECMWF. ERA5-Land hourly data. DOI: 10.24381/cds.e2161bac.
 11. U.S. Census Bureau, 2024 Gazetteer. ZCTA representative internal-point coordinates.
 12. NOAA/NCEI Access Data Service. Daily station summaries and request documentation.
 13. NOAA GHCN-Daily documentation. Variable definitions, observation attributes and quality flags.
 14. Open-Meteo daily aggregation discussion. Maintainer explanation of timezone behavior; saved response offsets control this analysis.
 15. statsmodels: `cov_nw_groupsum`. Time-aggregated Newey-West covariance reference implementation.
 16. statsmodels: generalized linear models. Poisson GLM numerical cross-check.
 17. statsmodels: `multipletests`. Holm family-wise multiple-testing adjustment.
 18. Sanz-Barbero B, Linares C, Vives-Cases C, González JL, López-Ossorio JJ, Díaz J. Heat wave and the risk of intimate partner violence. *Science of the Total Environment*. 2018;644:413-419. doi:10.1016/j.scitotenv.2018.06.368.
 19. Zhu Y, He C, Bell M, et al.. Association of Ambient Temperature With the Prevalence of Intimate Partner Violence Among Partnered Women in Low- and Middle-Income South Asian Countries. *JAMA Psychiatry*. 2023;80(9):952-961. doi:10.1001/jamapsychiatry.2023.1958.
 20. California Family Code 6211. Domestic-relationship definition, including specified family relationships.
 21. Study computational records, September 24, 2026. `audit.json`, `charge_classification_map.csv`, `model_results.csv`, `model_analysis_coverage.csv`, `noaa_crosscheck.csv`, `source_audit_pass2.json`, `weather_audit.md` and `statistical_audit/STATISTICAL_AUDIT.md` with result CSVs. See page 11.
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